

based on actual market conditions and aggregated buying and selling pressure from the open through 1:30 p.m. The bottom graph shows the corresponding (estimated) cumulative buy-sell imbalance in the stock over the same period of time. We used the I-Star model and corresponding parameters from Chapter 5 to compute these costs. The buy-sell imbalance was computed from tick data following the modified Lee & Ready tick rule. Then based on market conditions, volumes, volatility, and the buy-sell imbalance we are able to compute the trading cost. The graph shows that there was early buying pressure in the stock that pushed the trading cost up but the buying pressure was short lived. There was a subsequent sell-off in the stock beginning at around 10:15 a.m. continuing through about 1:15 p.m. causing prices to also decline.

The real-time cost index provides investors with a metric that can be used to measure the performance of their broker in real-time. Furthermore, this metric is transparent and can be verified by both parties, and computed over any specified time period used for the trading horizon. Brokers can no longer hide in the shadows and provide vague and elusive responses when questioned about their trading performance in real-time. The real-time cost index has leveled the playing field for buy-side and sell-side firms.

The most important aspect here is that it allows both broker and investor to better partner and determine when it is most appropriate to change intraday strategies and revise algorithmic trading parameters.

Figure 12.4 illustrates the end of day real-time cost index for AAPL during April 2012. The top graph shows the end of day real-time cost index given actual market conditions and buying-selling pressure. This is the cost an investor trading over the full day was expected to incur based on what actually occurred in the market conditions over that horizon. In actuality, the real-time cost index can be computed for any intraday time period. The bottom graph shows the estimated buy-sell imbalance on the day expressed as % ADV.

This metric can be used to further evaluate trader, broker, and algorithmic performance since it incorporates actual market activity. This measure can assess whether the VWAP price on the day was reasonable given market activity. Money managers have been using these metrics to determine whether there is any persistence of order imbalance in any stocks over time, which could signal a mispricing due to the market impact cost (here we have temporary impact disguised as permanent) with likely trend reversion. Money managers who uncover these signals could potentially earn a short-term profit.